Homework 2

PHYS 6020 - Fall 2023

Due on Friday, September 8th, 2023

Problem 1

Consider the wave-function of a free particle, in a one-dimensional problem at time $t = 0$ to be:

$$\psi(x, t = 0) = N \int_{-\infty}^{+\infty} dk e^{-|k|/k_0} e^{ikx} \tag{1}$$

where k_0 and N are constants.

a) What is the probability $\mathcal{P}(p_1, 0)$ that a measurement of the momentum performed at time $t = 0$ will yield a result included between $-p_1$ and $+p_1$?. Sketch the function $\mathcal{P}(p_1, 0)$.

b) What is the form of the wave-packet at time $t = 0$? Calculate for this time the product $\Delta X\, \Delta P$. What is your conclusion? Describe qualitatively the subsequent evolution of the wave-packet.

Hint: you may find useful to look over Section 1.8.3 in Zettili's book at the end of Chapter 1.

Problem 2

Consider now the three-dimensional wave-function:

$$\psi(x, y, z) = N e^{[-\frac{|x|}{2a} - \frac{|y|}{2b} - \frac{|z|}{2c}]} \tag{2}$$

where a, b, c are three positive lengths.

a) Calculate the constant N that normalizes ψ.

b) Calculate the probability that a measurement of $\hat{X}$ will yield a result included between 0 and a.

c) Calculate the probability that simultaneous measurements of $\hat{Y}$ and $\hat{Z}$ will yield results included respectively between $-b$ and b, and $-c$ and c.

d) Calculate the probability that a measurement of the momentum will yield a result included in the element $dp_x dp_y dp_z$ centered at the point $p_x = p_y = 0$ and $p_z = \hbar/c$.

For the following problems you may find useful to browse the various results presented in Chapter 1.

Problem 3

Solve problem 1.42 at the end of Chapter 1.

Problem 4

Solve problem 1.44 at the end of Chapter 1.

Problem 5

Solve problem 1.47 at the end of Chapter 1.